Differential Equations Workbook

Second order equations

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SECOND ORDER LINEAR HOMOGENEOUS EQUATIONS

■ 1. Find the general solution to the homogeneous equation.

$$y'' + 9y' + 8y = 0$$

■ 2. Find the solution to the homogeneous equation, if $y(0) = 5$ and $y'(0) = 2$.

$$y'' - 4y' + 5y = 0$$

■ 3. Find the general solution to the homogeneous equation.

$$y'' - 10y' + 25y = 0$$

■ 4. If $y(0) = 1$ and $y'(0) = 0$, find the solution to the homogeneous equation.

$$y'' + 4y' + 4y = 0$$

■ 5. If $y(0) = 2$ and $y'(0) = 3$, find the solution to the homogeneous equation.

$$y'' + 9y = 0$$

■ 6. If $y(0) = 1$ and $y'(0) = 5$, find the solution to the homogeneous equation.

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$$2y'' + 11y' + 5y = 0$$

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REDUCTION OF ORDER

■ 1. Use reduction of order to find the general solution to the differential equation, given $y_1 = e^{2x}$.

$$y'' + 3y' - 10y = 0$$

■ 2. Use reduction of order to find the general solution to the differential equation, given $y_1 = x^2$.

$$x^2y'' - 7xy' + 12y = 0$$

■ 3. Use reduction of order to find the general solution to the differential equation, given $y_1 = x^3$.

$$x^2y'' - 6y = x^3$$

■ 4. Use reduction of order to find the general solution to the differential equation, given $y_1 = \cos(2x)$.

$$y'' + 4y = 0$$

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■ 5. Use reduction of order to find the general solution to the differential equation, given $y_1 = x$.

$$(x+1)y'' + xy' - y = 0$$

■ 6. Use reduction of order to find the general solution to the differential equation, given $y_1 = e^{4x}$.

$$y'' - 5y' + 4y = 0$$

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UNDETERMINED COEFFICIENTS FOR NONHOMOGENEOUS EQUATIONS

■ 1. Find the general solution to the differential equation.

$$y'' - 2y' + y = x^3 - 6x^2 + 6x + 1$$

■ 2. Use the method of undetermined coefficient to solve the differential equation.

$$y'' - 3y' + 2y = e^{-x}(6x^2 - 10x + 20)$$

■ 3. Find the general solution to the differential equation.

$$y'' - y' = e^{2x} + e^{3x} + e^{4x}$$

■ 4. Solve the differential equation.

$$y'' + 9y = 12\sin(3x)$$

■ 5. Solve the differential equation.

$$y'' - 4y' + 4y = e^{2x}$$

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6. Solve the differential equation.

$$y'' - 4y' + 3y = e^x + 10\sin x + 3x$$

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VARIATION OF PARAMETERS FOR NONHOMOGENEOUS EQUATIONS

■ 1. Find the general solution to the differential equation.

$$y'' - 10y' + 25y = x^3 e^{5x}$$

■ 2. Find the general solution to the differential equation.

$$y'' - 6y' + 8y = e^{3x} + 5$$

■ 3. Find the general solution to the differential equation.

$$y'' - 7y' + 12y = \frac{e^{5x}}{e^{2x} + 1}$$

■ 4. Find the general solution to the differential equation.

$$y'' - 6y' + 9y = e^{3x} \sin(2x + 1)$$

■ 5. Find the general solution to the differential equation.

$$2y'' + 18y = \sin^4(3x)$$

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6. Find the general solution to the differential equation.

$$y'' - 4y' + 5y = e^{-2x} \ln(1 + \sin x)$$

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FUNDAMENTAL SOLUTION SETS AND THE WRONSKIAN

■ 1. Use the Wronskian to determine whether $\{y_1, y_2\} = \{e^x, e^{x+1}\}$ is the fundamental set of solutions for the differential equation.

$$y'' = y$$

■ 2. Determine whether $\{y_1, y_2\} = \{\cos(2x), \cos(2x+1)\}$ is a fundamental set of solutions for the differential equation.

$$y'' + 4y = 0$$

■ 3. Find the Wronskian for the solution set.

$$\{e^{3x}, xe^{3x}\}$$

■ 4. Find the Wronskian for the solution set.

$$\{e^{2x}\cos(2x), e^{2x}\sin(2x)\}$$

■ 5. Find the fundamental set of solutions for the the second order differential equation.

$$y'' + 2y' + y = 0$$

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■ 6. Find the fundamental set of solutions for the the second order differential equation, generated by solutions of the initial value problems with $y(0) = 1$ and $y'(0) = 0$, and $y(0) = 0$ and $y'(0) = 1$.

$$y'' + 4y' = 0$$

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VARIATION OF PARAMETERS WITH THE WRONSKIAN

■ 1. Solve the differential equation.

$$y'' + 7y' + 12y = e^x + 7e^{3x}$$

■ 2. Solve the differential equation.

$$y'' - 2y' = \frac{e^{3x}}{e^{2x} + 1}$$

■ 3. Solve the differential equation.

$$y'' + 8y' + 16y = 21x^5e^{-4x}$$

■ 4. Solve the differential equation.

$$y'' + 6y' + 9y = \frac{1}{e^{3x}\sqrt{1 - x^2}}$$

■ 5. Solve the differential equation.

$$y'' + y = \frac{1}{\sin^3 x}$$

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6. Solve the differential equation.

$$y'' + 4y = \cos^2(2x)$$

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INITIAL VALUE PROBLEMS WITH NONHOMOGENEOUS EQUATIONS

■ 1. Solve the initial value problem for the second order nonhomogeneous equation, given $y(0) = 0$ and $y'(0) = 2$.

$$y'' - 2y' - 3y = 10e^x \sin x$$

■ 2. Solve the initial value problem for the second order nonhomogeneous equation, given $y(0) = 0$ and $y'(0) = 0$.

$$y'' - 4y = 8x^2 + 4$$

■ 3. Solve the initial value problem for the second order nonhomogeneous equation, given $y(0) = 1$ and $y'(0) = 0$.

$$y'' - 10y' + 25y = 28x^6 e^{5x}$$

■ 4. Solve the initial value problem for the second order nonhomogeneous equation, given $y(0) = 1$ and $y'(0) = 0$.

$$4y'' + 4y' + y = 25 \sin x + 4$$

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■ 5. Solve the initial value problem for the second order nonhomogeneous equation, given $y(0) = 0$ and $y'(0) = 1$.

$$y'' + 4y' + 5y = 17e^{2x} + 40\sin(3x)$$

■ 6. Solve the initial value problem for the second order nonhomogeneous equation, given $y(0) = 2$ and $y'(0) = 2$.

$$y'' + 16y = 40\sin^2(3x)$$

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